ITA0604-MACHINE LEARNING

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Backpropagation Example: Predicting Access to Sanitation Facilities (SDG 6 – Clean Water and Sanitation)

Problem Statement

Design a simple neural network that predicts whether a **slum household has access to sanitation facilities** based on two input features.

Inputs

- **x_1 = Household Income (normalized)** = 0.60
- **x_2 = Education Level (normalized)** = 0.80

Output

- **y = 1** (Household has access to sanitation)

Network Architecture

x1 = 0.60

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H1 ------\

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x2 = 0.80 Output

(Sanitation Access)

- Input Layer: 2 neurons
- Hidden Layer: 1 neuron
- Output Layer: 1 neuron
- Activation Function: Sigmoid

- Learning Rate (η): 0.1

Initial Weights

Input → Hidden

- $w_1 = 0.20$
- $w_2 = 0.40$
- Hidden Bias = 0.10

Hidden → Output

- $w_3 = 0.50$
- Output Bias = 0.20

Step 1: Forward Pass

Hidden Layer

Net input

$$\text{��}_h = \text{��}_1\text{��}_1 + \text{��}_2\text{��}_2 + \text{��}_h$$
$$= (0.6)(0.2) + (0.8)(0.4) + 0.1$$
$$= 0.12 + 0.32 + 0.10$$
$$= 0.54$$

Hidden Activation

Sigmoid

$$h = \frac{1}{1 + \text{��}^{-0.54}}$$
$$h = 0.632$$

Output Layer

Net input

$$\text{��}_{\text{��}} = h \times \text{��}_3 + \text{��}_{\text{��}}$$
$$= (0.632)(0.5) + 0.2$$

$$= 0.516$$

Output Activation

$$\hat{y} = \frac{1}{1 + e^{-0.516}}$$

$$\hat{y} = 0.626$$

Prediction

Predicted Output

$$\boxed{0.626}$$

Actual Output

$$\boxed{1}$$

Step 2: Error Calculation

Mean Squared Error

$$E = \frac{1}{2}(y - \hat{y})^2$$

$$= \frac{1}{2}(1 - 0.626)^2$$

$$= \frac{1}{2}(0.374)^2$$

$$E = 0.0699$$

Step 3: Backward Pass

Output Layer Delta

$$\delta_{o} = (y - \hat{y})\hat{y}(1 - \hat{y})$$

$$= (1 - 0.626)(0.626)(1 - 0.626)$$

$$= 0.0875$$

Update Hidden → Output Weight

$$\Delta ��_3 = ����_{��} h$$
$$= (0.1)(0.0875)(0.632)$$
$$= 0.00553$$

New Weight

$$��_3 = 0.50 + 0.00553$$
$$\boxed{��_3 = 0.5055}$$

Update Output Bias

$$\Delta ��_{��} = ����_{��}$$
$$= (0.1)(0.0875)$$
$$= 0.00875$$

New Bias

$$\boxed{��_{��} = 0.2088}$$

Hidden Layer Delta

$$��_h = h(1 - h)(��_{��}��_3)$$
$$= (0.632)(1 - 0.632)(0.0875)(0.5)$$
$$= 0.0102$$

Update Input → Hidden Weights

Weight w_1

$$\Delta ��_1 = ����_h ��_1$$
$$= (0.1)(0.0102)(0.6)$$
$$= 0.00061$$

New Weight

$$\boxed{��_1 = 0.2006}$$

Weight w_2

$$\Delta ��_2 = ����_h ��_2$$
$$= (0.1)(0.0102)(0.8)$$
$$= 0.00082$$

New Weight

$$\boxed{��_2 = 0.4008}$$

Update Hidden Bias

$$\Delta ��_h = ����_h$$
$$= (0.1)(0.0102)$$
$$= 0.00102$$

New Bias

$$\boxed{��_h = 0.1010}$$

Updated Parameters After One Training

Iteration Parameter Old Value New Value

w_1 0.2000 **0.2006**

w_2 0.4000 **0.4008**

Hidden Bias 0.1000 **0.1010**

w_3 0.5000 **0.5055**

Output Bias 0.2000 **0.2088**

Flow Diagram

Forward Pass

Income (0.6) ──┐
├──► Hidden Neuron (0.632)

Education(0.8)┘ │

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Output = 0.626

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Error = 0.0699

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Backpropagation

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Update Weights and Biases

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Improved Prediction

Conclusion

The forward pass computes a sanitation access prediction of **0.626** for the household. The backward pass calculates the prediction error and updates the network's weights and biases using gradient descent. Repeating this process over many training examples reduces the error, enabling the model to accurately predict whether households have access to sanitation facilities, supporting **SDG 6 – Clean Water and Sanitation**.